

NATIONAL JUNIOR COLLEGE, SINGAPORE
Senior High 2
Preliminary Examination
Higher 2

CANDIDATE
NAME

BIOLOGY
CLASS

REGISTRATION
NUMBER

Biology

9744/03

Paper 3 Long Structured and Free-response Questions

28 August 2018

2 hours

Additional Materials: Answer Paper

READ THESE INSTRUCTIONS FIRST

Write your name, Biology class and registration number on all the work you hand in.
Write in dark blue or black pen.
You may use an HB pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Sections A and B

Answer **all** questions in the spaces provided on the Question Paper.

Section C

Answer any **one** question on the answer paper provided.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in the brackets [] at the end of each question or part of question.

For Examiner's Use	
Section A	
1	/30

This document consists of **16** printed pages.

Section A

Answer the question in this section.

- 1 (a) Chinese white poplar (*Populus tomentosa*) is a native tree species in China that grows in temperate areas. Its growth-dormancy cycle is driven by environmental cues such as temperature.

In a study in the early spring, all the mRNA from the cells in the cambium region of *P. tomentosa* were isolated on six different days in 2008 and converted to complementary DNA (cDNA). The cDNA was then used for polymerase chain reaction (PCR) analysis of the expression patterns of five cell cycle-related genes (*PtoCKS1*, *PtoCYCD3*, *PtoCYCB*, *PtoCDKB* and *PtoCDKA*) and one housekeeping gene (*UBQ-L*)

Fig. 1.1 shows the expression patterns of the six genes in the cambium region of *P. tomentosa*.

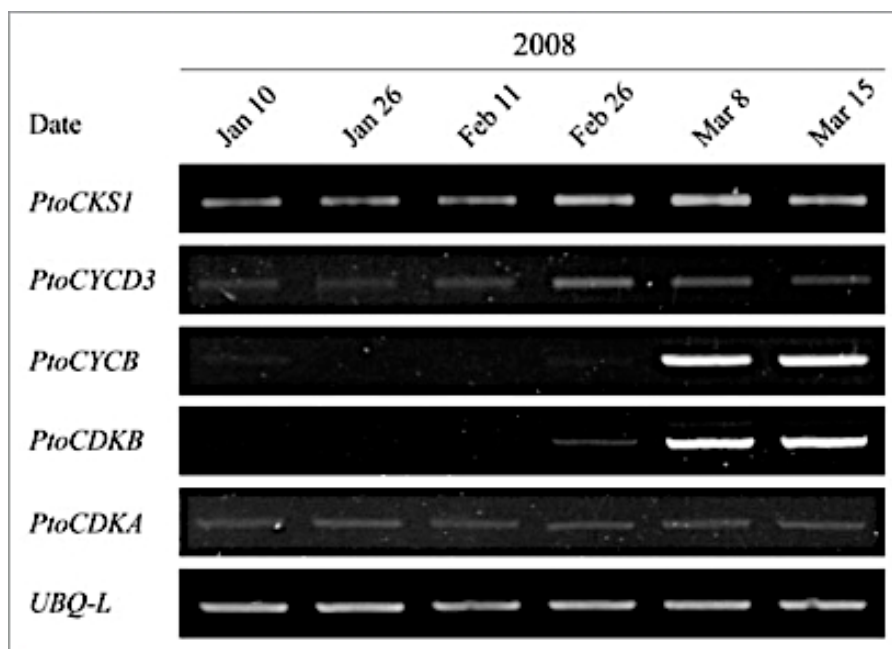


Fig. 1.1

- (i) Outline the process of PCR.

[3]

- (ii) With reference to Fig. 1.1, describe the expression patterns of *PtoCKS1*, *PtoCYCB*, and *UBQ-L*.

[3]

- (iii) During the shift from dormancy to growth, the mitotic cell cycle occurs at an accelerated rate in the cells of *P. tomentosa*.

Outline the behaviour of chromosomes in mitosis.

[4]

- (iv) Other than growth, explain **two** other functions of mitosis.

[2]

- (b) The increase in temperature affects leaf metabolism in several ways. It affects the activity of Rubisco, the central enzyme of photosynthesis, as well as Rubisco activase, the enzyme required for Rubisco reactivation. In addition, Rubisco activase is found to be far more sensitive to heat than Rubisco.

Fig. 1.2 shows the effect of two temperatures on the activity of the two enzymes.

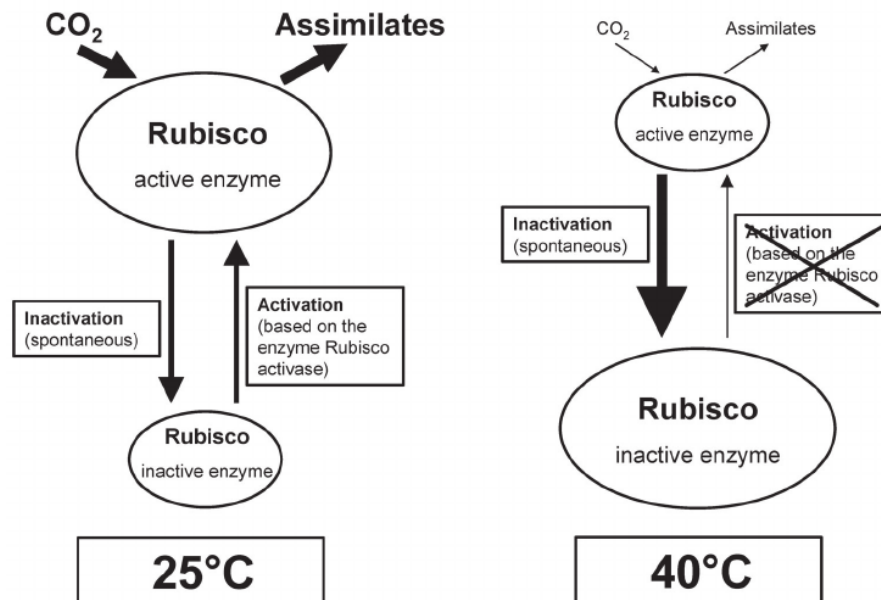


Fig. 1.2

- (i) Suggest **one** reason why Rubisco activase is far more sensitive to heat than Rubisco.

[1]

- (ii) Describe the effect of increasing temperature from 25°C to 40°C on Rubisco activase activity.

[4]

- (iii) With reference to Fig. 1.2, describe and explain the difference between the rate of photosynthesis at 25°C and that at 40°C.

[3]

- (c) Other than temperature, increasing CO₂ concentration could also affect the rate of photosynthesis.

- (i) Identify and explain **one** human activity over the last few centuries that has contributed to an increase in atmospheric CO₂ concentration.

[2]

- (ii) Explain the effects of climate change as atmospheric CO₂ concentration increases.

[3]

- (d) Dietary deficiencies of zinc and iron are a substantial global public health problem. An estimated two billion people worldwide suffer from these deficiencies. Most of these people depend on C3 grains and legumes as their primary dietary source of zinc and iron.

A study was done to determine the concentration of zinc when C3 rice plants were grown in a controlled environment at the current atmospheric CO₂ concentration, and at the elevated atmospheric CO₂ concentration predicted for the middle of this century.

Table 1.1 shows the results of the study.

Table 1.1

concentration of zinc in C3 rice plants / mg kg ⁻¹	
grown at the current atmospheric CO ₂ concentration	grown at the elevated atmospheric CO ₂ concentration predicted for 2050
80	75
62	55
52	50
70	62
64	58
72	55
69	70
70	62

A two-sample t-test could be carried out to determine whether there is a decrease in the concentration of zinc when the C3 rice plants were grown at the elevated atmospheric CO₂ concentration predicted for 2050.

- (i) State the alternative hypothesis for the t-test.

[1]

- (ii) Carry out the t-test in the space provided.
Show all your working.

[2]

- (iii) State the conclusion for the t-test.

[2]

[Total: 30]

This page is intended to be blank.

Name: _____

Biology Class: 2bi2_____

Section B

For Examiner's Use	
Section B	
2	/20

Section B

Answer the question in this section.

- 2** *Helicobacter pylori* is a spiral shaped, gram-negative bacterium that lives in the stomach and duodenum. In Singapore, the prevalence rate in the community (without any symptom) is estimated to increase with age — from 3% in children under the age of 5 to 71% in adults above the age of 65.

H. pylori has a genome size of approximately 1.7 Mb, with a G+C content of 35 % to 40%. Its genome sequence is highly variable. The genome of strain 26695 includes 1,587 genes, whereas the genome of strain J99 includes only 1,491 genes. Many strains carry one or more plasmids, which do not seem to carry antibiotic resistance genes or virulence genes.

- (a)** Compare the structure and organisation of *H. pylori* genome with its host cell genome.

[4]

Fig. 2.1 shows the major virulence and colonization factors of *H. pylori*. *H. pylori* uses its flagella and chemotaxis mechanisms to swim through the mucus lining the gastric epithelium. It attaches to the host cell surface and deliver toxins, such as CagA and VacA, into the host cell to control various aspects of host cell function.

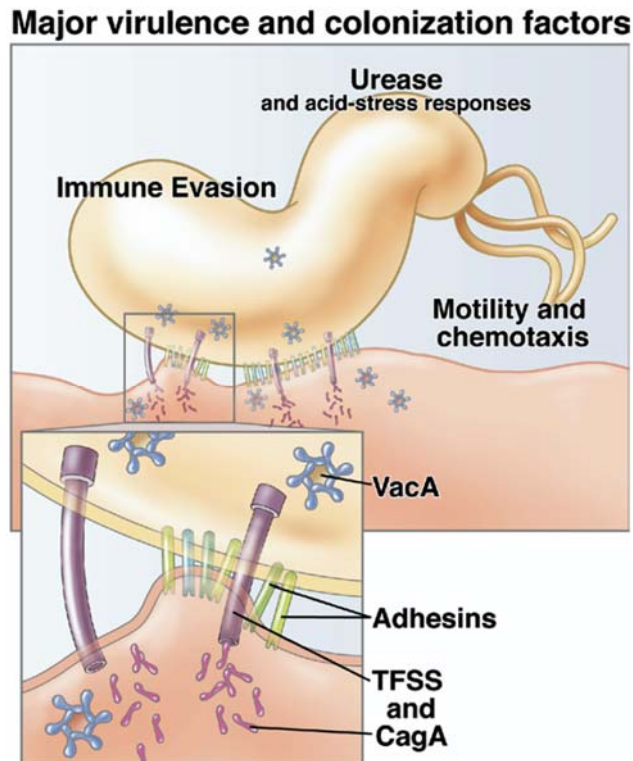


Fig. 2.1

- (b) There are several ways to diagnose *H. pylori*. One method is to draw a sample of blood from the patient and test the blood for presence of *H. pylori*-specific antibody.

Explain how the presence *H. pylori* triggers the production of *H. pylori*-specific antibodies in an infected patient.

[3]

The first line therapy for *H. pylori* infection is known as triple therapy, comprising two antibiotics, clarithromycin and amoxicillin, and one proton pump inhibitor. All three drugs are administered two times a day for 14 days. Amoxicillin is similar in action to penicillin while clarithromycin binds to 23S rRNA.

- (c) Discuss the deliberate use of the two antibiotics in the therapy.

[4]

Early studies found that patients with antibodies against CagA showed higher rates of gastric adenocarcinoma. The C-terminal tail of the CagA is a target for phosphorylation by kinases coded by cellular oncogenes. After phosphorylation, CagA would trigger signals that resemble the activation of receptor tyrosine kinase by growth factors, controlling proliferative activities, adhesion, and cytoskeletal organization of epithelial cells. Other studies have shown that CagA leads to increased degradation of p53.

- (d) Using the information provided, explain why patients with prolonged *H. pylori* infection have an increased risk of developing adenocarcinoma.

[4]

cagA is a gene that has many variants. In particular, there are different numbers of repeat sequences located in the 3' region of the *cagA* gene of different *H. pylori* strains. Each repeat region of the CagA protein contains Glu-Pro-Ile-Tyr-Ala (EPIYA) motifs, including a tyrosine phosphorylation site. Studies confirmed that, in western countries, the incidence of gastric cancer is notably higher in patients infected with strains containing multiple EPIYA-C segments than in patients infected with strains containing a single EPIYA-C segment.

Fig. 2.2 shows three variants of CagA protein.

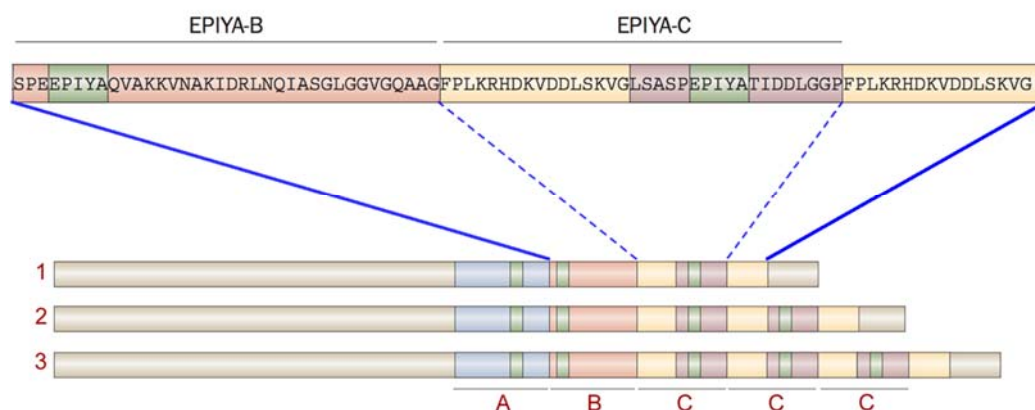


Fig. 2.2

- (e) (i) Polymerase chain reaction (PCR) could be carried out on a tissue sample to determine the *cagA* variant of the *H. pylori* strain infecting the patient.

Indicate on Fig. 2.2, the ideal position of **one** set of primers to distinguish the DNA sequence corresponding to the three CagA variants by PCR.

[1]

- (ii) Determine the size difference between PCR products from tissue samples containing variant **1** and variant **2** respectively.

[1]

- (iii) Outline the principle and procedure of a molecular technique that could be used to determine the variant of *cagA* gene after PCR has been performed.

[3]

[Total: 20]

Name: _____

Biology Class: 2bi2_____

Section C

For Examiner's Use	
Section C	
___ (a)	/15
___ (b)	/10

Section C

Answer **one** question in this section.

Write your answers on the separate answer paper provided.

Your answers should be illustrated with large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)**, as indicated in the question.

- 3 (a)** Using **three** named examples, explain the relationship between the structures and functions of carbohydrates in living organisms. [15]
- (b)** Describe how photosynthesis and cellular respiration complement each other in the environment, and compare the two processes. [10]

[Total: 25]

- 4 (a)** Using **three** named examples, explain the relationship between the structures and functions of proteins in living organisms. [15]
- (b)** Describe how enzymes increase the rate of chemical reactions, and compare the actions of competitive and non-competitive enzyme inhibitors. [10]

[Total: 25]